Janet Gaba

7141 Sherbrooke St. W., Montreal, QC, Canada H4B 1R6 · janet.gaba@mail.concordia.ca

EDUCATION

Ph.D., Chemistry 2026
Concordia University, Montreal, QC, Canada

Dissertation title:
Electronic Structure Effects on Phenolic Surfactant Self-Assembly and Peptide Binding

GPA: 4.30

Scholarships received:
Hydro-Quebec Doctoral Scholarship, 2020-2021, $11,160 CAD
R. Howard Webster Foundation Doctoral Fellowship, 2020-2021, $10,000 CAD

Certificate received:
Graduate Seminar in University Teaching, 2021

Award received:
Outstanding Teaching Assistant Award, Department of Chemistry and Biochemistry, 2023

B.Sc., Biochemistry 2017
Concordia University, Montreal, QC, Canada

Coursework, Biochemistry 2004-2008
Otterbein University, Westerville, OH, USA

Honours:
Alpha Lambda Delta, Phi Eta Sigma, Dean's list (3 quarters)

RESEARCH EXPERIENCE

Postdoctoral Fellow June 2026 - present
Concordia University, Montreal, QC, Canada
Principal Investigator: Prof. Heidi Muchall

Studying noncovalent interactions driving the assembly of phenolic surfactant clusters, in order to better understand gallate surfactant monolayer behaviours and direct future syntheses of surfactants. Performing delocalization index analysis within the Quantum Theory of Atoms in Molecules (QTAIM) on gallate-analogue geometries calculated using density-functional theory (DFT). Preparing final manuscript from doctoral research for submission. Assisting with supervision and mentorship of graduate students doing computational research projects.

Doctoral Research Assistant Jan. 2018 - Dec. 2025
Concordia University, Montreal, QC, Canada
Principal Investigators: Profs. Heidi Muchall & Christine DeWolf

Investigated causes of experimentally-observed behavior of gallate surfactant monolayers, for future use as biosensing coatings for devices. Identified and assessed electronic effects on monolayer self-assembly, organization, and binding to peptide partners. Techniques used include electronic structure (DFT and semiempirical) methods, *e.g.* energy minimization, frequency, and natural bond orbital calculations, and electron density analysis using QTAIM; surface pressure-area isotherms; and grazing incidence X-ray diffraction experiments (performed at Argonne National Laboratory). Supervised undergraduate specialization and honours research projects.

Undergraduate Research Assistant
Concordia University, Montreal, QC, Canada Sept. 2016 - Dec. 2016
Principal Investigator: Prof. Heidi Muchall

Studied changes in ^{17}O chemical shifts upon intramolecular hydrogen bonding in phenols and benzaldehydes, using electronic structure methods. Evaluated validity of past experiments.

Undergraduate Research Assistant
Otterbein University, Westerville, OH, USA Feb. 2005 - June 2008
Principal Investigator: Prof. Joseph Sachleben

Analyzed effects of paramagnetism and crystallinity on the NMR spectrum by comparing solid-state ^{13}C NMR spectra of $Sm(acac)_3$, $Y(acac)_3$, and mixed crystals thereof. Performed simulations of ^{13}C NMR rotational decoupling sidebands to compare to experimental data.

PUBLICATIONS

Peer-Reviewed Articles

Sachleben, J. R.; **Gaba, J. H.**; Emsley, L. Floquet-van Vleck Analysis of Heteronuclear Spin Decoupling in Solids: The Effect of Spinning and Decoupling Sidebands on the Spectrum. *Solid State Nucl. Mag.* **2006**, *29*, 30-51. https://doi.org/10.1016/j.ssnmr.2005.09.009

Gaba, J. H.; DeWolf, C. E.; Muchall, H. M. Noncovalent interactions governing self-assembly of oligomeric phenolic surfactant film models. *J. Phys. Org. Chem*. **2026**, *39*, e70073. https://doi.org/10.1002/poc.70073

Gaba, J. H.; DeWolf, C. E.; Muchall, H. M. Electronic effects on the self-assembly of monolayers from surfactants possessing aromatic headgroups. *Langmuir* **2026**, *42*, 5977-5984. https://doi.org/10.1021/acs.langmuir.5c03809

In Preparation

Miclette Lamarche, R.*; **Gaba, J. H.***; Muchall, H. M.; DeWolf, C. E. Effects of proline on the assembly and organization of phenolic surfactant monolayers. Submitting to *Colloids Surf. A* end of July 2026.

** Co-first authors*

CONFERENCE PRESENTATIONS

Oral Presentations

Gaba, J. H.; DeWolf, C. E.; Muchall, H. M. Weak interactions in the binding of poly-L-proline and gallate monolayers. Presented at the 50th Ontario-Quebec Physical Organic Mini-Symposium, Kingston, ON, November 5, 2022.

Gaba, J. H.; DeWolf, C. E.; Muchall, H. M. Noncovalent interactions in models of surfactant monolayers. Presented at the 2nd Centre for Research in Molecular Modeling Symposium, Montreal, QC, May 12, 2022.

Gaba, J. H.; DeWolf, C. E.; Muchall, H. M. Toward a protein-responsive coating: a computational study of interactions of phenolic surfactant monolayers with poly-L-proline. Presented at the 104th Canadian Chemistry Conference and Exhibition, remote, August 13-20, 2021.

Gaba, J. H.; DeWolf, C. E.; Muchall, H. M. Effects of π-π Stacking on Film Formation of Phenolic Surfactant Monolayers. Presented at the 1st Centre for Research in Molecular Modeling Symposium, Montreal, QC, February 7, 2020.

Gaba, J. H.; DeWolf, C. E.; Muchall, H. M. π-π Stacking in Models of Phenolic Surfactant Monolayers. Presented at the 22nd Chemistry and Biochemistry Graduate Research Conference, Montreal, QC, November 15, 2019.

Gaba, J. H.; DeWolf, C. E.; Muchall, H. M. π-π Stacking as a Driving Force in Film Formation at the Air-Water Interface. Presented at the 47th Ontario-Quebec Physical Organic Mini-Symposium, St. Catharines, ON, November 1-3, 2019.

Gaba, J. H.; DeWolf, C. E.; Muchall, H. M. Computational Study of Weak Interactions in Assemblies of Phenolic Surfactants. Presented at the 102nd Canadian Chemistry Conference and Exhibition, Quebec, QC, June 3-7, 2019.

Poster Presentations

Gaba, J. H.; DeWolf, C. E.; Muchall, H. Effects of proline on the assembly and organization of phenolic surfactant monolayers. Presented at the 53rd Ontario-Quebec Physical Organic Mini-Symposium, Ottawa, ON, November 15, 2025.

Owino, J.; Yu, Z.; **Gaba, J. H.**; DeWolf, C. E.; Muchall, H. M. Toward patterning of phenolic surfactant monolayers: a computational study. Presented at the 108th Canadian Chemistry Conference and Exhibition, Ottawa, ON, June 18, 2025.

Gaba, J. H.; DeWolf, C. E.; Muchall, H. M. Electronic effects on the organization of surfactants with aromatic headgroups in self-assembled monolayers. Presented at the Canadian Institute for Chemistry Montreal Section Annual Symposium, Montreal, QC, May 26, 2025.

Owino, J.; **Gaba, J. H.**; Muchall, H. M. A Computational Study on the Origins of Stacking and Displacement in Phenolic Surfactant Models. Presented at the 5th Centre for Research in Molecular Modeling Symposium, Montreal, QC, May 6, 2025.

<u>Gaba, J. H.</u>; DeWolf, C. E.; Muchall, H. M. Influence of electronic effects on the self-assembly of monolayers from surfactants with aromatic headgroups. Presented at the 5th Centre for Research in Molecular Modeling Symposium, Montreal, QC, May 6, 2025.

<u>Krome, A. K.</u>; **Gaba, J. H.**; DeWolf, C. E.; Muchall, H. A Computational Investigation of the Intermolecular Interactions Driving the Dimerization of Stacked Para-Substituted Phenols. Presented at the 52nd Ontario-Quebec Physical Organic Mini-Symposium, Montreal, QC, November 2, 2024.

<u>Gaba, J. H.</u>; DeWolf, C. E.; Muchall, H. Phenolic surfactants: from molecules to monolayers. Presented at the 51st Ontario-Quebec Physical Organic Mini-Symposium, Kingston, ON, November 4, 2023.

<u>Gaba, J. H.</u>; DeWolf, C. E.; Muchall, H. M. Computational investigation of interactions in models of phenolic surfactant monolayers binding with poly-L-proline. Presented at the 7th Biophysical Society of Canada Meeting, Ottawa, ON, May 25, 2022.

<u>Gaba, J. H.</u>; DeWolf, C. E.; Muchall, H. M. Computational Study of Weak Interactions in Cyclic and π-stacked Arrangements of Model Phenolic Surfactants. Presented at the 21st Chemistry and Biochemistry Graduate Research Conference, Montreal, QC, November 9, 2018.

<u>Gaba, J. H.</u>; DeWolf, C. E.; Muchall, H. M. Computational Study of Weak Interactions in Cyclic Trimers of Model Phenolic Surfactants. Presented at the 46th Ontario-Quebec Physical Organic Mini-Symposium, St. Catharines, ON, November 2-4, 2018.

<u>Gaba, J. H.</u>; DeWolf, C. E.; Muchall, H. M. Computational Study of Weak Interactions in Small Assemblies of Model Phenolic Surfactants. Presented at the 36th Chemistry Graduate Student Symposium of SUNY Buffalo, Buffalo, NY, May 21-23, 2018.

<u>Gaba, J. H.</u>; Muchall, H. M. Computational Study into the Origin of Chemical Shift Changes upon Formation of Weak Interactions. Presented at the 44th Ontario-Quebec Physical Organic Mini-Symposium, St. Catharines, ON, November 4-5, 2016.

<u>Gaba, J. H.</u>; Sachleben, J. R. Solid-State NMR of Lanthanide Acetylacetonates. Presented at the 40th Central Regional Meeting of the American Chemical Society, Columbus, OH, June 10-14, 2008.

<u>Gaba, J. H.</u>; Sachleben, J. R. Solid-State NMR Characterization of Lanthanide Acetylacetonates. Presented at the 115th Annual Meeting of the Ohio Academy of Science, Dayton, OH, April 21-23, 2006.

TEACHING EXPERIENCE

Graduate Teaching Assistant Jan. 2018 - Dec. 2025
Concordia University, Montreal, QC

General Chemistry I (CHEM 205)

- Tutorials: led weekly to twice-weekly sessions of 20-40 students each. Prepared materials for question & answer sessions, prepared problem sets, led students in solving problems, administered and marked quizzes, and addressed questions during office hours.

- In-person labs: Led weekly laboratory sessions of 20-25 students. Demonstrated proper use of equipment, ensured safety, answered questions, marked reports, held office hours.
- Remote labs: Led weekly remote laboratory sessions over Zoom; students did kitchen chemistry using mailed kits and items from local shops. Demonstrated proper use of equipment, ensured safety, answered questions, marked reports, held office hours.
- Administration: Assisted in administration of the CHEM 205 laboratory, which had more than 500 students across 11 lab sections. Answered questions on course structure, collated attendance records, facilitated movement between sections for make-up sessions, assisted in training of new TAs, helped solve issues with experiments over Zoom.

General Chemistry II (CHEM 206)

- Tutorials: Led weekly to twice-weekly sessions of 20-30 students each. Prepared materials for question & answer sessions, led students in solving problems, administered and marked quizzes, and addressed questions during office hours.
- Remote labs: Led weekly remote laboratory sessions via Zoom; students performed simulations of experiments using the virtual laboratory software Stemble. Presented lab topics, answered questions, marked reports, held office hours.

Thermodynamics (CHEM 234)

- Tutorials: Led sessions of 10-30 students, four times weekly. Topics included calculus review, corrections to the ideal gas law, thermochemistry, the laws of thermodynamics, Helmholtz and Gibbs energies, phase transitions, activities, and mixtures. Prepared supplementary materials, led students in solving problems, and answered questions.

Inorganic Chemistry I (CHEM 241)

- Computational laboratory marker: Marked reports and assisted students at office hours with topics including atomic and molecular orbitals and modelling.

Organic Spectroscopy (CHEM 293)

- Tutorials: Debuted organic spectroscopy tutorial sessions at Concordia University: 10-20 students each, three to four times weekly, five weeks per term. Conducted tutorials remotely using Zoom and Moodle (2020-2021), and in person (2023-2024).

  Wrote and presented a course prerequisite quiz, and three-hour lessons on determination of molecular formulas; infrared (IR) and ultraviolet-visible (UV-Vis) spectroscopies; simple ^{1}H nuclear magnetic resonance (NMR) spectra; complex multiplets, ^{13}C NMR spectra, and two-dimensional NMR techniques; and mass spectrometry.

  Lessons included experimental techniques, theory, prediction, and interpretation; a pre-tutorial reading assignment; a group problem-solving exercise on a real-world topic; and an individual quiz. Recorded videos to show later TAs how to present these topics.
- Laboratory: Led weekly laboratory sessions of 10-20 students. Answered student questions, performed demonstrations, and aided student experiments using techniques including IR (with and without attenuated total reflectance), UV-Vis, and ^{1}H NMR spectroscopies. Taught computational methods to predict and explain data.

Biochemistry II (CHEM 375)

- Laboratory marker: Marked reports and assisted students at office hours with topics including enzyme assays and computational modelling of proteins.

Undergraduate Teaching Assistant Sept. 2006 - Nov. 2007
Otterbein University, Westerville, OH

Assisted with LSCI 101 (first-term biology) laboratory sections. Answered questions on calculations, laboratory technique (*e.g.* dilutions, light microscopy, and gel electrophoresis), and cellular structure. Ensured safety, assisted with demonstrations, and fixed equipment.

SERVICE

Presentation Session Chair, CBGRC Novembers, 2018 - 2025

Chaired physical and computational chemistry oral presentation sessions for the Chemistry and Biochemistry Graduate Research Conference at Concordia University.

Councillor, Arts and Science Federation of Associations June 2016 - June 2017

- Spoke and voted on issues of student interest during council meetings of the Arts and Science Federation of Associations (ASFA), representing constituent undergraduate students in the Department of Chemistry and Biochemistry and the Department of Physics.
- Became familiar with Robert's Rules of Order.
- On the Finance Committee, voted on and tracked the disbursement of $200,000 to 28 member associations for their operating budgets.

KEY SKILLS

Chemistry · Molecular modelling · Data analysis · Teaching & tutoring · Verbal & written communication · Problem solving · Leadership & collaboration · Adaptability & flexibility · Moodle · Gaussian · Python · R · Wolfram Mathematica · Bash · English (native) · French

---

REFERENCES

Prof. Heidi Muchall 1 (514) 848-2424 #3342 heidi.muchall@concordia.ca
Associate Professor, Department of Chemistry and Biochemistry, Concordia University

Prof. Christine DeWolf 1 (514) 848-2424 #4225 christine.dewolf@concordia.ca
Professor, Department of Chemistry and Biochemistry, Concordia University

Prof. Gregor Kos 1 (514) 848-2424 #3374 gregor.kos@concordia.ca
Senior Lecturer, Department of Chemistry and Biochemistry, Concordia University